

The Triumph of Simplicity

How simple services are reshaping databases and the enterprise

“Life is really simple, but we insist on making it complicated.”

-Confucius

Who am I?

HELLO,
MY NAME
IS IAN PLOSKER

Co-founder & CTO



@dstroyallmodels

about.me/ian.plosker

Orchestrate.io is a service that combines the best of NoSQL into a simple HTTP-based service
Previously, I worked for Basho, the makers of Riak

Our goal is to make storing and
querying so easy, you don't need
databases



E.W. Dijkstra

“Simplicity is a great virtue but it requires hard work to achieve it and education to appreciate it. And to make matters worse: complexity sells better.”

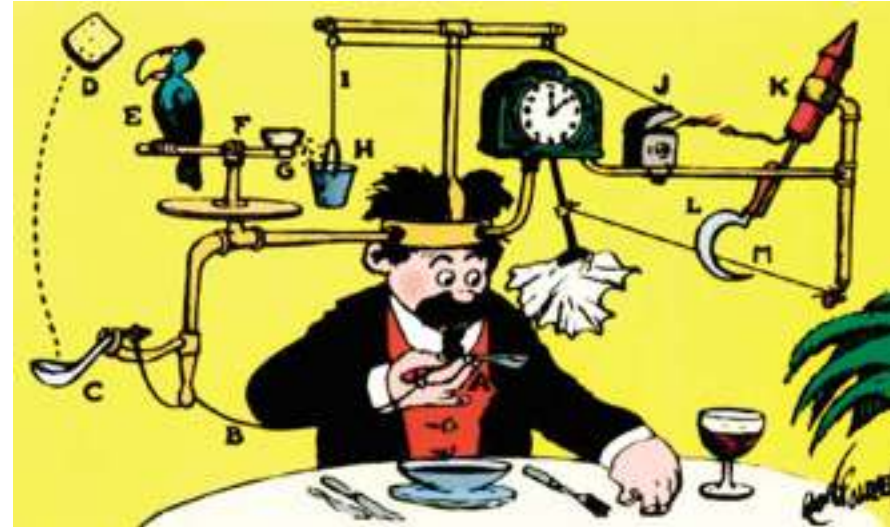
<http://www.cs.utexas.edu/users/EWD/transcriptions/EWD08xx/EWD896.html>

Dijkstra Algorithm
Semaphores
Proponent of Structured programming
Read some of Dijkstra's speeches, they are indictments of the computing industry



Some people believe
that complexity is how
value is added

We fetishize complexity



<http://www.cs.utexas.edu/users/EWD/transcriptions/EWD08xx/EWD898.html>

“... many a programmer derives a major part of his professional excitement from not quite understanding what he is doing, from the daring risks he takes and from the struggle to find the bugs he should not have introduced in the first place.” -Dijkstra

Increasingly, people seem to misinterpret complexity as sophistication, which is baffling --- the incomprehensible should cause suspicion rather than admiration. Possibly this trend results from a mistaken belief that using a somewhat mysterious device confers an aura of power on the user -- Niklaus Wirth (Pascal)

Some guy got his rocks off over this query

There is a quiet revolution in
progress

1. open source
2. cloud
3. apis



Ronald Coase

“... a firm will tend to expand until the costs of organizing an extra transaction within the firm become equal to the costs of carrying out the same transaction by means of an exchange on the open market or the costs of organizing in another firm.”

<http://www3.nccu.edu.tw/~jsfeng/CPECI1.pdf>

Nobel Prize in Economics
Died at 102, last month (2/9/13)
The Nature of the Firm (1937)

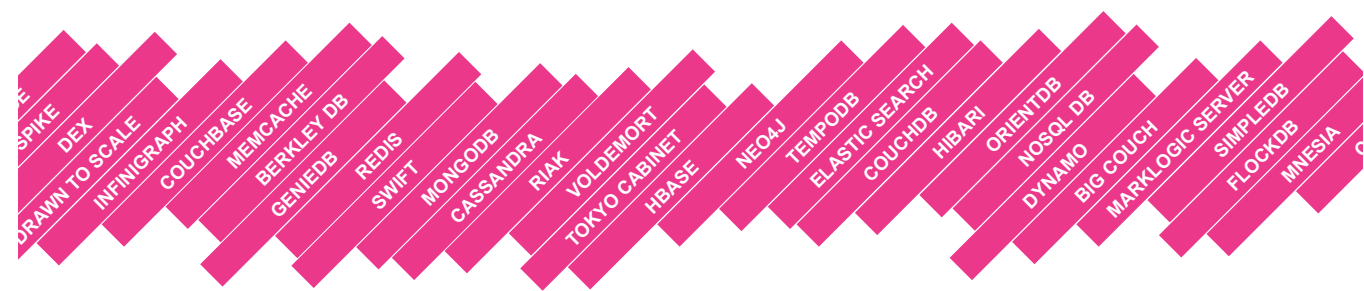
Functions that organizations
formerly did themselves are now
sourced externally

Let's talk databases

Databases in 2005



Databases in 2013



25 databases in production today that didn't exist 7 years ago

How many people run a NoSQL DB?
How many are thinking about it?
How many would like to?
Cassie? MongoDB? Riak?

ONLINE QUERY TYPES

	Key-Value	Search	Geo	Graph/ Relation	Event
<i>Scale-up</i>	BerkleyDB CouchDB MongoDB MySQL	SOLR Sphinx	PostGIS MongoDB SOLR	neo4j	MySQL
<i>Scale-out</i>	Riak Cassandra	elasticsearch	elasticsearch	titan	HBase

The database paradox of choice:



Choice has brought complexity

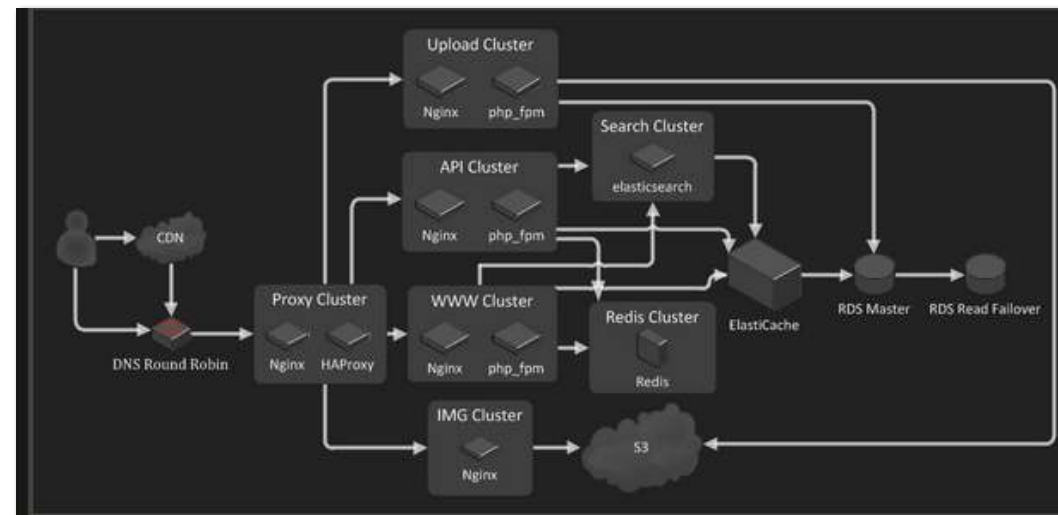
Klout's Data Architecture



1. Hbase
2. MySQL
3. ElasticSearch
4. MongoDB



Imgur's Data Architecture



1. Hbase 2. ElasticSearch 3. MySQL 4. Memcache 5. Redis 6. HAproxy

Who Owns Complexity?

“... making something “simpler” is often a case of relocating complexity, rather than eliminating it from the user-technology relationship. For example, from the driver’s perspective, a manual-shift transmission is more complex than an automatic transmission. But from an overall systems perspective, the automatic transmission is equally or even more complex.”

Simple



Complex



Reiterate the car example

Simple

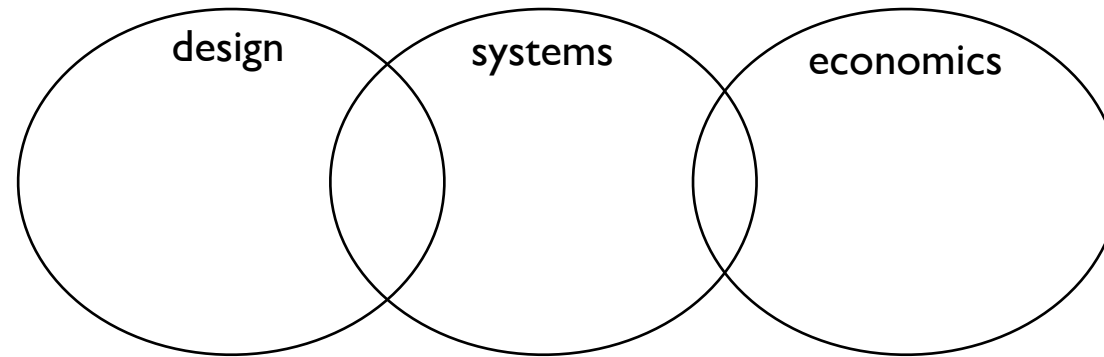


Complex

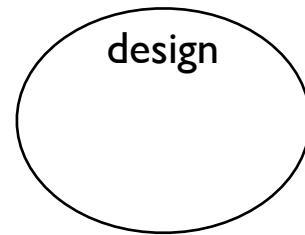


The ultimate in simplicity of interface vs complexity of systems

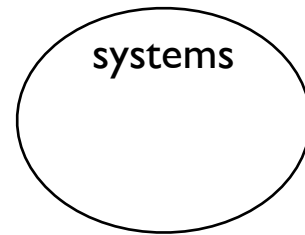
The Value of Simplicity Varies Across Domains



Simple or complex?



Simple or complex?



Simple or complex?

economics

Net Digital Ad Revenue Share worldwide						
	2011		2012		2013	
	Share (percent)	Revenue (\$ bn)	Share (percent)	Revenue (\$ bn)	Share (percent)	Revenue (\$ bn)
Google	32.08	27.72	31.46	32.73	32.84	38.62
Facebook	3.65	3.15	4.11	4.28	5.41	6.36
Yahoo!	3.95	3.41	3.37	3.51	2.97	3.50
Microsoft	2.60	2.25	2.46	2.56	2.49	2.92
IAC	1.01	0.87	1.26	1.32	1.37	1.62
AOL	1.17	1.01	1.02	1.06	0.94	1.11
Amazon	0.48	0.42	0.59	0.61	0.71	0.84
Twitter	0.16	0.24	0.28	0.38	0.50	0.58
Pandora	0.28	0.14	0.36	0.29	0.49	0.58
LinkedIn	0.18	0.16	0.25	0.26	0.32	0.38
Millennial Media	0.05	0.04	0.07	0.07	0.10	0.12
Other	54.40	47.02	54.77	56.98	51.85	60.97
Total (USD in billions)	\$86.43		\$104.04		\$117.60	

Source: eMarketer

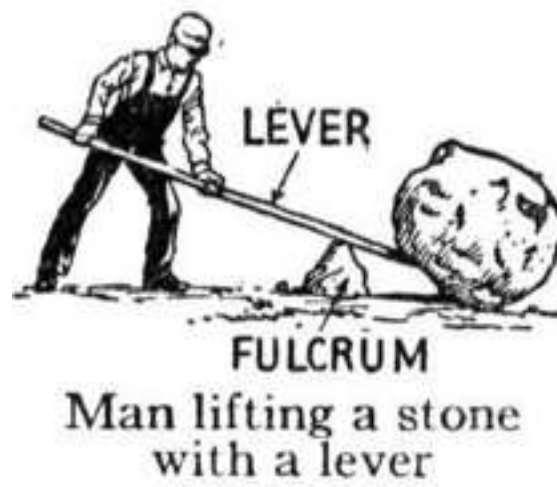
Design -> Systems -> Economics Alignment is low for Google
 It s high for Amazon
 It s moderate for Apple
 It s high for LinkedIn
 It s low for Twitter

↑
API

expose utility, hide complexity

The other overlap

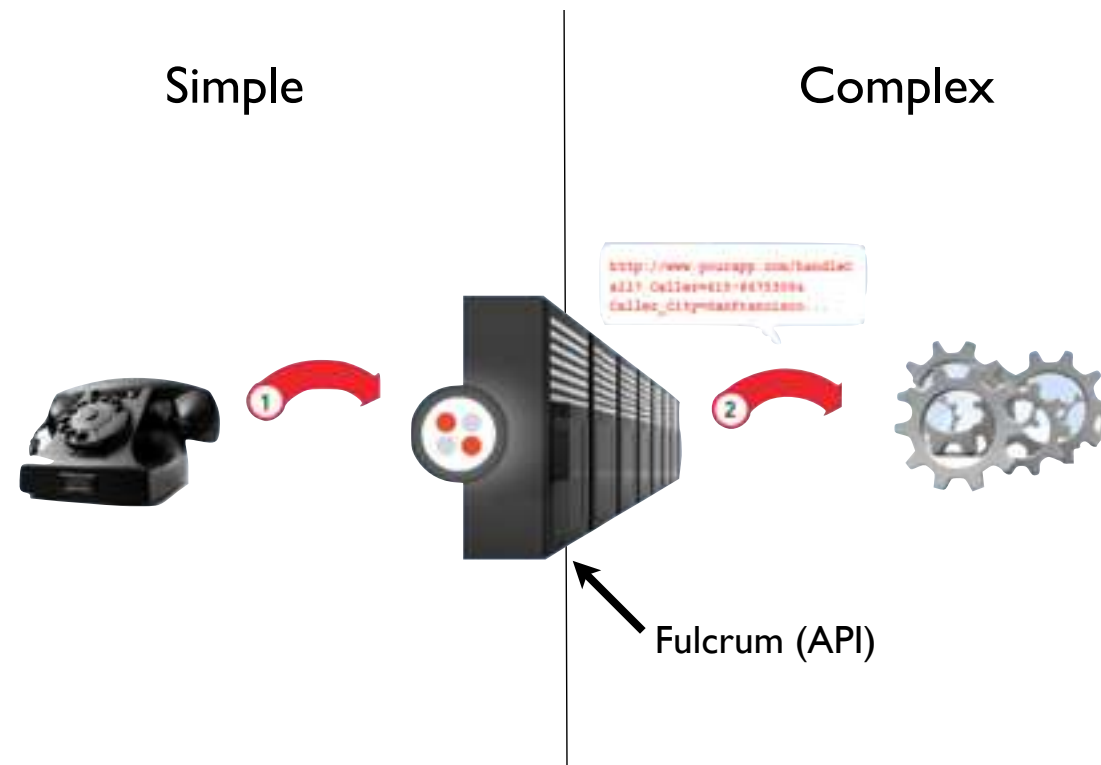
Introducing: Leverage



when a complex system has a simple interface

good simplicity is complexity disguised, not complexity denied

The Complexity of Simplicity, Kris Jordan, Dec 4, 2009, <http://www.newmediacampaigns.com/page/simple-design-is-complex>



Twilio shifts the headache and complexity of systems to a 3rd party

Simple

Complex



PayPal™

stripe



Simple

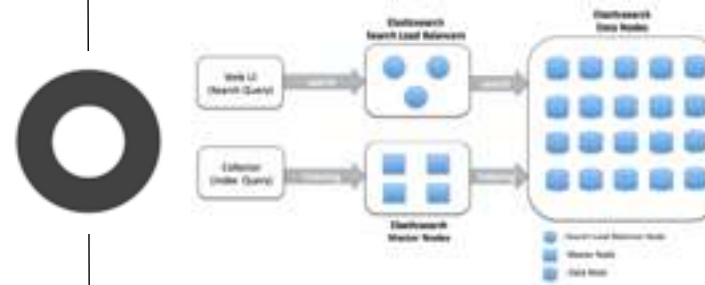


Complex



Simple

Complex



This doesn't represent the supporting infrastructure (monitoring, metrics collection, log aggregation, request tracing, etc)

Simple

Complex



This doesn't represent the supporting infrastructure (monitoring, metrics collection, log aggregation, request tracing, etc)

Orchestrate.io is a HTTP REST
service that unifies 5 common
query types

Key/Value

Full-text Search

Graph

Time-ordered events

Geolocation

LIVE DEMO!!!!!!

Simplicity

h/t @AntonyFalco

Simplicity is arriving at a conclusion that in retrospect, appears inevitable.

-Francis Pedreza

Simple things should be simple and complex things should be possible. -- Alan Kay (OO)

Simplicity and elegance are unpopular because they require hard work and discipline to achieve and education to be appreciated.

-- Edsger W. Dijkstra